NBP

27/11/2024

Code A Phase-1

Aakash
Medical|IIT-JEE|Foundations

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MM : 720 NCERT Booster Test Series for NEET (OYM)-2024-25_Test-01 **Time : 200 Min.**

Topics Covered:
Physics : Thermodynamics, Kinetic Theory, Oscillations, Waves, Electromagnetic Induction
Chemistry : Organic Chemistry: Some Basic Principles and Techniques, Hydrocarbons, Solutions, Electrochemistry
Botany : Photosynthesis in Higher Plants, Respiration in Plants, Plant Growth and Development, Sexual Reproduction in Flowering Plants
Zoology : Animal Kingdom : General Accounts, Non-chordates & chordates, Human Health and Disease

General Instructions :

1. There are two sections in each subject, i.e. **Section-A & Section-B**. You have to attempt all **35 questions** from Section-A & only **10 questions** from Section-B out of **15.**
2. Each question carries **+4 marks**. For every wrong response, **–1 mark** shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
3. Use blue/black ballpoint pen only to darken the appropriate circle.
4. Mark should be dark and completely fill the circle.
5. Dark only one circle for each entry.
6. Dark the circle in the space provided only.
7. Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

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PHYSICS

SECTION-A

1. Read the following statements.
(a) Magnetic flux is a vector quantity
(b) The SI unit of magnetic flux is T m^{-2}
(c) The dimensional formula for magnetic flux is $[ML^2T^{-2}A^{-1}]$
The incorrect statement(s) is/are

(1) (a) only

(2) (b) only

(3) (a) and (b) only

(4) All (a), (b) and (c)

2. **Statement I:** If magnetic flux through a fixed loop decreases, then magnetic field due to induced current will be opposite to the existing magnetic field.
Statement II: A bar magnet is falling along the axis of circular loop. Its acceleration will be less than g after it crosses the plane of the loop.

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

3. The magnetic flux across a loop of resistance 100 Ω is given by $\phi = (2t^2 - 2t)$ Wb. The current induced in the loop at $t = 2$ s is

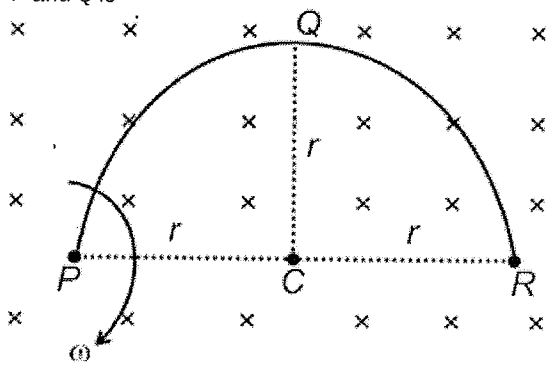
(1) 40 mA

(2) 60 mA

(3) 80 mA

(4) 120 mA

4. A semicircular loop of radius r is rotated in a plane about one of its ends as shown. The magnetic field (B) exists perpendicular to the plane of loop. If ω is angular speed of rotation, then magnitude of induced emf between the points P and Q is



(1) $\frac{1}{2} B\omega r^2$

(2) $B\omega r^2$

(3) $\frac{B\omega r^2}{4}$

(4) $2B\omega r^2$

5. A conducting square loop of constant side length L and resistance R is moving inside transverse magnetic field B with constant speed v. The power required to do so is

(1) $\frac{B^2L^2v}{R}$

(2) $\frac{B^2L^2v^2}{R}$

(3) Zero

(4) $\frac{B^2L^2v}{2R}$

6. When a particle is executing SHM with frequency f, then potential energy of the particle

(1) Varies periodically with frequency f

(2) Varies periodically with frequency $f/2$

(3) Varies periodically with frequency $f/4$

(4) Varies periodically with frequency $2f$

7. A body oscillates in SHM according to equation $y = 5\sin\left(2\pi t + \frac{\pi}{4}\right)$ m. Its instantaneous speed of body at t = 1 s (in m s^{-1}) is

(1) $5\sqrt{2}\pi$

(2) $\frac{5}{\sqrt{2}}\pi$

(3) 5π

(4) 10π

8. A block of mass 1 kg is attached to a spring of spring constant 100 N m^{-1}. The block is displaced by 10 cm form its equilibrium position and released. The minimum time taken by block to reach mean position is

(1) $\frac{\pi}{10}$ sec

(2) $\frac{\pi}{20}$ sec

(3) $\frac{\pi}{30}$ sec

(4) $\frac{\pi}{40}$ sec

9. **Assertion (A):** If a simple pendulum is taken to a lift accelerating upwards with acceleration (a), then its time period of oscillation will decrease.
Reason (R): The time period of oscillation for a pendulum in the lift accelerating upward with acceleration a is $T' = 2\pi\sqrt{\frac{l}{g+a}}$

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) Only (A) is true but (R) is false

(4) (A) is false but (R) is true

10. Out of the following functions representing motion of the particle, which function(s) represent SHM?
(a) $y = \sin\omega t + \cos\omega t$
(b) $y = \sin\omega t - \cos 2\omega t$
(c) $y = \sin^2\omega t + \cos^2\omega t$
(d) $y = \cos^2\omega t - \sin^2\omega t$

(1) (a) only

(2) (b) and (d) only

(3) (a) and (d) only

(4) All (a), (b), (c) and (d)

11. Consider the following two statements
a. On a humid day, sound in air, travels slower than on a dry day
b. Sound waves cannot propagate in vacuum.
The correct statement is

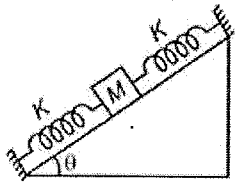
(1) a only

(2) b only

(3) a and b both

(4) Neither a nor b

12. If the inclined plane is smooth and the block is attached with ideal springs as shown. Then time period of the block for small oscillations is



(1) $2\pi\sqrt{\frac{M}{2k}}$

(2) $2\pi\sqrt{\frac{M\sin\theta}{2k}}$

(3) $2\pi\sqrt{\frac{2M}{k}}$

(4) $2\pi\sqrt{\frac{2M}{k\sin\theta}}$

13. The equation of S.H.M. of a particle whose amplitude is 2 m and frequency 50 Hz, which starts from extreme position is

(1) $y = 2\sin(100\pi t)$ m

(2) $y = 2\sin(50\pi t)$ m

(3) $y = 2\cos(50\pi t)$ m

(4) $y = 2\cos(100\pi t)$ m

14. A tuning fork of unknown frequency produces 8 beats/s with a tuning fork of frequency 212 Hz. The latter fork is now waxed and the beat frequency reduces to 4 beats per second. The unknown frequency is

(1) 211 Hz

(2) 215 Hz

(3) 204 Hz

(4) 220 Hz

15. A 10 m long steel wire has mass of 5 g. This wire is under a tension of 80 N. The speed of transverse wave travelling on it is

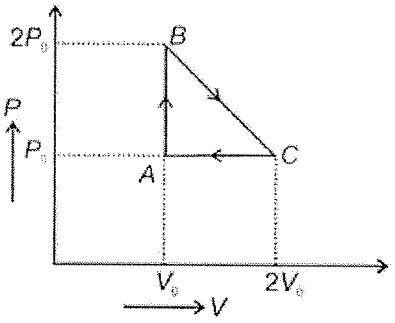
(1) 200 m s^{-1}

(2) 300 m s^{-1}

(3) 340 m s^{-1}

(4) 400 m s^{-1}

16. Two moles of an ideal diatomic gas is taken around the cyclic process *ABCA* as shown in pressure (*P*)-volume (*V*) diagram. The total work done by the gas in process *ABCA* will be



(1) $\frac{-5P_0V_0}{2}$

(2) $\frac{-P_0V_0}{2}$

(3) $\frac{P_0V_0}{2}$

(4) Zero

17. If the frequency of a sonometer wire is to be increased to double its value, then the tension in the wire should be increased by

(1) 50%

(2) 100%

(3) 200%

(4) 300%

18. Two sound waves of same frequency travel in the same direction in a medium. The amplitude of each wave is *A* and phase difference between the two waves is $\frac{2\pi}{3}$. The amplitude of resultant wave formed after their superposition is

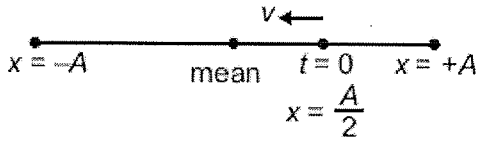
(1) *A*

(2) 2*A*

(3) $\sqrt{2A}$

(4) $\frac{A}{2}$

19. A particle performing simple harmonic motion is shown in the figure below. The correct equation for SHM is (symbols have their usual meanings)



(1) $x = A\sin\left(\omega t + \frac{3\pi}{2}\right)$

(2) $x = A\sin\left(\omega t + \frac{\pi}{6}\right)$

(3) $x = A\sin\left(\omega t + \frac{\pi}{2}\right)$

(4) $x = A\sin\left(\omega t + \frac{5\pi}{6}\right)$

20. The phase difference between displacement and acceleration of simple harmonic oscillator is

(1) π

(2) $\frac{\pi}{2}$

(3) $\frac{\pi}{4}$

(4) $\frac{\pi}{3}$

21. If a mixture of equal mass of N_2 and O_2 are kept in a container in thermal equilibrium, then the gases have same

(1) Root mean square speed of gas molecules

(2) Internal energy

(3) Number of molecules

(4) Internal energy per mole

22. Choose the correct statement.

(1) Internal energy is a path function while heat is not

(2) Heat is a path function while internal energy is not

(3) Both heat and internal energy are not path functions

(4) Both heat and internal energy are path functions

23. If volume of gas is decreased by 20%, then change in pressure will be (temperature remains constant)

(1) 15% increase

(2) 10% increase

(3) 25% increase

(4) 20% increase

24. The efficiency of heat engine working between 27°C and 327°C may be

(1) 50%

(2) 35%

(3) 25%

(4) all of these

25. A progressive wave of frequency 500 Hz is travelling with a velocity of 360 m/s. The distance between two points 60° out of phase is

(1) 12 cm

(2) 6 cm

(3) 10 cm

(4) 2 cm

26. The root mean square speed of the molecules of a gas is 900 m/s. The most probable speed of the molecules is about

(1) 800 m/s

(2) 734.8 m/s

(3) 302 m/s

(4) 534 m/s

27. A bulb contains one mole of oxygen at temperature 300 K. Another bulb contains one mole helium at temperature 200 K. The ratio of rms speed of helium to the most probable speed of oxygen is

(1) 2 : 1

(2) 4

(3) $2\sqrt{2} : 1$

(4) $\sqrt{2} : 1$

28. The ratio of mean free paths of two gas molecules with molecular radii r and $2r$ respectively is (Assume number density of gases is same)

(1) 4 : 1

(2) 2 : 1

(3) 1 : 2

(4) 4 : 3

29. A particle of mass 4 kg is moving along x-axis under the action of force $F = -\pi^2 x$ (where F is in newton and x is in m). The time period of oscillation is

(1) 1 s

(2) 2 s

(3) 3 s

(4) 4 s

30. A block of mass m connected with a spring of constant K is performing S.H.M. with time period T. If spring is cut into two parts of length $l_1 : l_2 = 2 : 3$ and same block is connected with smaller section of spring instead, then time period will be

(1) $\frac{T}{\sqrt{3}}$

(2) $\frac{T}{\sqrt{5}}$

(3) $\sqrt{\frac{3}{2}}T$

(4) $\sqrt{\frac{2}{5}}T$

31. The magnetic flux of 500 μWb per turn passing through a 200 turn coil reversed in 20×10^{-3} s. The average emf induced in the coil in volt is

(1) 2.5

(2) 7.5

(3) 5.0

(4) 10.0

32. The rate of increase of current through a inductor is 2.4 A/s. If voltage drop across the inductor is 1.6 V then self inductance of the inductor will be

(1) $\frac{2}{3}$ H

(2) $\frac{1}{3}$ H

(3) $\frac{3}{2}$ H

(4) $\frac{1}{2}$ H

33. A : If there is change in magnetic flux, an emf is induced.
R : A current must be induced due to change in flux.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

34. The total charge flown through a conducting loop in a given time duration when it is moved in magnetic field

(1) Does not depend on magnetic flux

(2) Depends on initial magnetic flux only

(3) Depends on the total change in magnetic flux

(4) Depends on final magnetic flux only

35. The temperature at which root mean square speed of the molecules of a gas will be twice of its value at 27°C is

(1) 1200°C

(2) 600 K

(3) 927°C

(4) 927 K

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SECTION-B

36. The equation of a stationary wave in fundamental mode is given by $y = 0.4 \sin(4\pi x) \cos(5\pi t)$, where x, y are in meter and t is in seconds. Match the following columns and choose the correct option.

	Column I		Column II
A.	Amplitude of constituent wave	(P)	0.28 m
B.	Position of displacement node	(Q)	0.25 m
C.	Position of displacement antinode	(R)	0.2 m
D.	Displacement (y) of particle at $x = \frac{1}{16}$ m and $t = 0$	(S)	0.125 m

(1) A(R), B(S), C(Q), D(P)

(2) A(R), B(Q), C(S), D(P)

(3) A(S), B(Q), C(R), D(P)

(4) A(P), B(R), C(Q), D(S)

37. Equation of travelling wave is given by $y = A\sin(kx - \omega t)$. The ratio of maximum speed of the particle to the wave speed is

(1) 1

(2) $\frac{\omega}{k}$

(3) $A\omega$

(4) Ak

38. **Assertion (A):** Speed (v) of sound in air remains constant with increase in pressure.
Reason (R): The increase in pressure (P) increases air density(ρ). So, $v = \sqrt{\frac{\gamma P}{\rho}} = \text{constant}$

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) Only (A) is true but (R) is false

(4) (A) is false but (R) is true

39. A wave of frequency 300 Hz has a wave velocity 300 ms^{-1}. The phase difference between two points which are 0.25 m apart (in rad) is

(1) $\frac{\pi}{4}$

(2) $\frac{\pi}{3}$

(3) $\frac{\pi}{2}$

(4) π

40. A sound wave is passing through an air column in the form of compression and rarefactions. In the consecutive compressions and rarefactions,

(1) displacement of medium particles is maximum at both the positions.

(2) displacement of medium particles is zero at both the positions.

(3) Pressure change is zero at both the positions.

(4) Pressure change is zero at compression and non-zero at rarefactions.

41. In a resonance column, two successive resonances are obtained at 35 cm and 49 cm, respectively. The next resonance will be obtained at depth of

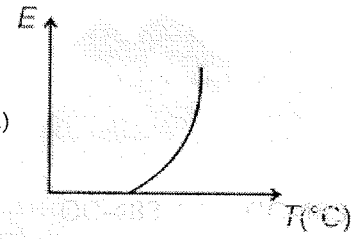
(1) 56 cm

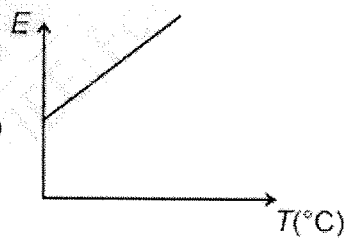
(2) 63 cm

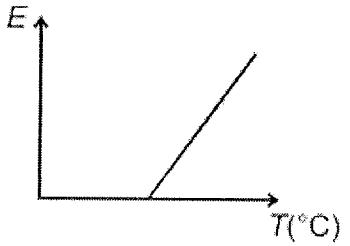
(3) 70 cm

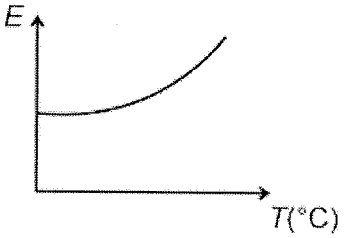
(4) 52.5 cm

42. The graph which represents the variation of mean kinetic energy (E) of molecules with temperature (T) in °C varies as

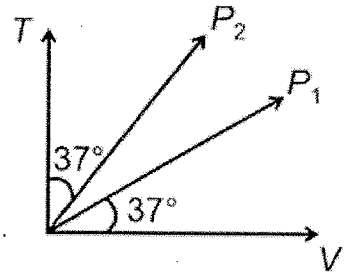
(1)


(2)


(3)


(4)


43. In the adjacent T-V diagram, the ratio $\frac{P_1}{P_2} =$



(1) $\frac{16}{9}$

(2) $\frac{9}{16}$

(3) $\frac{4}{3}$

(4) $\frac{3}{4}$

44. A gaseous mixture consists of 4 mol of O_2 and 4 mol of Ar at temperature T. Neglecting all vibrational modes, the total internal energy of the system is

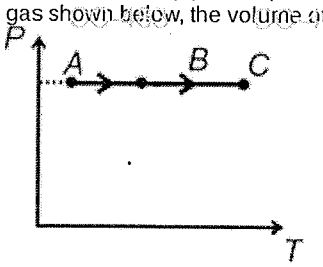
(1) $4RT$

(2) $8RT$

(3) $12RT$

(4) $16RT$

45. In the pressure (P) – temperature (T) diagram of an ideal gas shown below, the volume of the gas



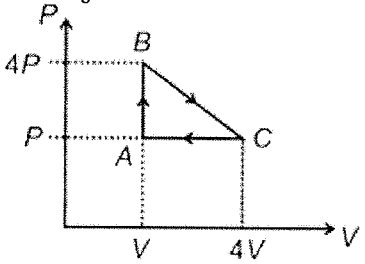
(1) is constant for process A to C

(2) Increases from A to B and then decreases from B to C

(3) Decreases from A to B and then increases from B to C

(4) Increases continuously from A to C

46. An ideal gas is taken around cycle ABCA as shown in the P-V diagram.



The work done in the process BC is

(1) $\frac{13}{2}PV$

(2) $\frac{15}{2}PV$

(3) $\frac{11}{2}PV$

(4) $7PV$

47. In a process for an ideal gas, $\Delta W = 0$ and $\Delta U < 0$. Then for the process

(1) Volume of gas decreases

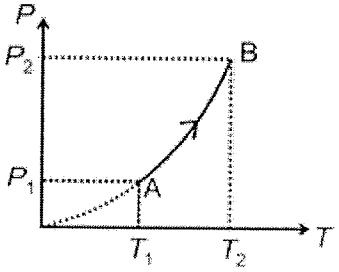
(2) Temperature of gas increases

(3) Pressure of gas increases

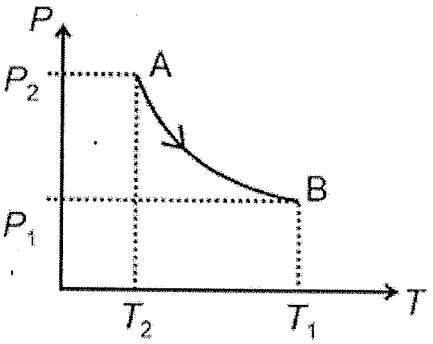
(4) Heat is released by the gas

48. An ideal monoatomic gas undergoes adiabatic process (AB). Pressure-temperature (P-T) graph for this process is

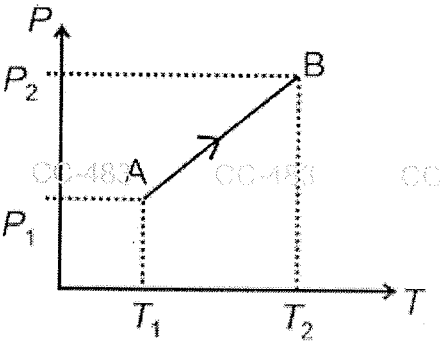
(1)



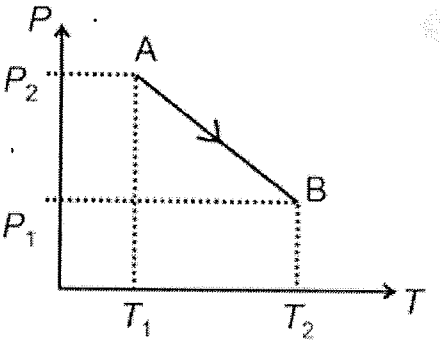
(2)



(3)



(4)



49. The displacement of two particles of same mass executing SHM are represented by the equations

$x_1 = 2 \sin\left(10t + \frac{\pi}{4}\right)$ and $x_2 = 4 \sin(\omega t)$

The value of ω for which both have same maximum energy is equal to

(1) 4

(2) 5

(3) $\frac{5}{2}$

(4) 2

50. Progressive wave through a stretched wire is given by $y = 5\sin(100\pi t - 0.4\ \pi x)$ where x, y are in meter and t is in second. If the tension in the rope is 625 N, then mass per unit length of the wire is

(1) 10^{-2} kg/m

(2) 10^{-3} kg/m

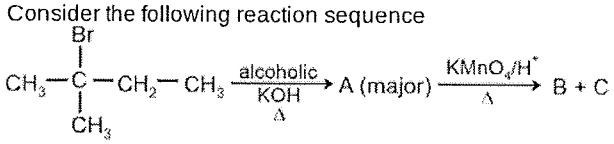
(3) 10^{-4} kg/m

(4) 10^{-1} kg/m

CHEMISTRY

SECTION-A

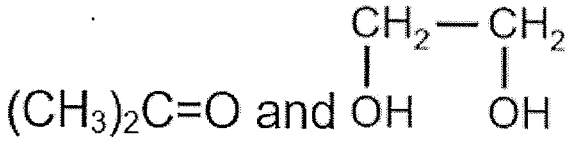
51. Consider the following reaction sequence



Major products B and C are

(1) $(CH_3)_2CH–OH$ and $CH_3CH_2–OH$

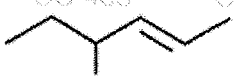
(2) $(CH_3)_2C=O$ and CH_3COOH

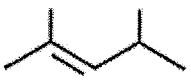
(3) 

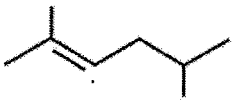
(4) CH_3CH_2COOH and CH_3COOH

52. Which among the following compounds on reductive ozonolysis gives isobutyraldehyde as one of the products?

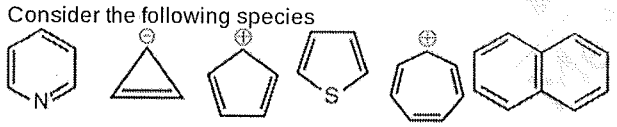
(1) 

(2) 

(3) 

(4) 

53. Consider the following species



(i) (ii) (iii) (iv) (v) (vi)

The number of aromatic species are

(1) 2

(2) 3

(3) 4

(4) 5

54. Given below are the two statements

Statement I: Melting point of neopentane is higher than hexane.

Statement II: Boiling point of isopentane is higher than neopentane

In the light of above statements, choose the correct answer

(1) Statement I is correct but statement II is incorrect

(2) Both statement I and statement II are correct

(3) Statement I is incorrect but statement II is correct

(4) Both statement I and statement II are incorrect

55. In sulphur estimation, 0.75 g of an organic compound gave 1.398 g of barium sulphate. The percentage of sulphur present in the compound is (Atomic mass of Ba = 137 u)

(1) 17.2%

(2) 42.8%

(3) 32.5%

(4) 25.6%

56. Consider the following statements.

(a) In positive electromeric effect, the π-electrons of multiple bond are transferred to that atom to which reagent gets attached.

(b) Electromeric effect is a permanent effect

(c) Benzyl carbocation does not show hyperconjugation effect.

(d) Hyperconjugation involves delocalisation of σ electrons of C–H bond of an alkyl group directly attached to an atom of unsaturated system or to an atom with an unshared p-orbital.

The correct statements are

(1) (a), (c) and (d) only

(2) (a), (b) and (c) only

(3) (b), (c) and (d) only

(4) (a), (b), (c) and (d)

57. Kjeldahl method is not applicable to which pair of compounds?

(1) Acetonitrile and Acetamide

(2) Aniline and Triethylamine

(3) Pyridine and Nitrobenzene

(4) Benzylamine and Allylamine

58. Which compound on reaction with sodium, liberates hydrogen gas?

(1) $CH_2=CH_2$

(2) $CH_3-C\equiv C-CH_3$

(3) $CH_3-C\equiv CH$

(4) $CH_3-CH_2-CH_2-CH_2-CH_3$

59. Number of σ and π bonds present in a phenanthrene molecule respectively are

(1) 28 and 7

(2) 19 and 5

(3) 24 and 6

(4) 26 and 7

60. In Carius method of estimation of halogen, 1.25 g of an organic compound gave 1.128 g of AgBr. The percentage of bromine present in the compound is (Molar mass of AgBr = 188 g mol^{-1})

(1) 38.4%

(2) 27.2%

(3) 19.6%

(4) 48.1%

61. Consider the following binary solutions of different compounds
(a) Bromoethane and chloroethane
(b) Ethanol and acetone
(c) Chloroform and acetone
(d) Phenol and aniline
(e) Carbon disulphide and acetone
The solutions which show positive deviation from Raoult's law are

(1) (c) and (d) only

(2) (b) and (e) only

(3) (a), (b) and (e) only

(4) (b), (c) and (d) only

62. If mass percent $\left(\frac{w}{w}\right)$ of methanol in water is 24%, then molality of methanol in the solution will be

(1) 7.5 m

(2) 9.9 m

(3) 6.8 m

(4) 5.1 m

63. If 6 g of urea is present in 400 mL of its aqueous solution at 27°C. The osmotic pressure of the solution will be (R = 0.083 L bar mol^{-1} K^{-1})

(1) 6.2 bar

(2) 3.8 bar

(3) 1.7 bar

(4) 9.2 bar

64. If 0.2 molal aqueous solution of sodium sulphate undergoes 90% dissociation, then the freezing point of the solution will be (K_f of water = 1.86 K kg mol^{-1})

(1) – 2.08°C

(2) – 1.5°C

(3) – 0.5°C

(4) – 1.04°C

65. Consider the following statements about dry cell
(a) The cell consists of zinc container that also acts as anode
(b) Electrolyte used is a paste of KOH and ZnO
(c) The cathode is graphite rod surrounded by powdered manganese dioxide and carbon.
The correct statement are

(1) (a) and (b) only

(2) (b) and (c) only

(3) (a), (b) and (c)

(4) (a) and (c) only

66. Correct order of limiting molar conductivity of given ions in water at 298 K is

(1) $H^+ > OH^- > Cl^- > Na^+$

(2) $H^+ > Na^+ > OH^- > Cl^-$

(3) $OH^- > H^+ > Na^+ > Cl^-$

(4) $OH^- > Cl^- > H^+ > Na^+$

67. The standard Gibbs energy of the given cell reaction will be
$3Ni(s) + 2Au^{3+}(aq) \rightarrow 3Ni^{2+}(aq) + 2Au(s)$
(Given: $E^\circ_{Ni^{2+}/Ni} = -0.25$ V and $E^\circ_{Au^{3+}/Au} = 1.4$ V)

(1) – 477.67 kJ mol^{-1}

(2) – 955.35 kJ mol^{-1}

(3) – 318.45 kJ mol^{-1}

(4) – 636.9 kJ mol^{-1}

68. If a solution of copper sulphate on electrolysis for 100 minutes deposits 3.86 g of copper at cathode then the current passed through the solution was (Atomic mass of copper = 63.5 u)

(1) 3.25 A

(2) 1.95 A

(3) 4.25 A

(4) 2.75 A

69. If the conductivity of 0.5 M NaBr solution is 0.045 S cm^{-1} then the molar conductivity of the solution will be

(1) 120 S cm^2mol^{-1}

(2) 100 S cm^2mol^{-1}

(3) 150 S cm^2mol^{-1}

(4) 90 S cm^2mol^{-1}

70. Mass of concentrated nitric acid solution required to prepare 800 mL of 4.0 M HNO_3 solution is (Concentrated HNO_3 is 70% $(\frac{w}{w})$ HNO_3)

(1) 288 g

(2) 340 g

(3) 460 g

(4) 510 g

71. Which among the following is most reactive towards electrophilic substitution reaction?

(1) 

(2) 

(3) 

(4) 

72. The correct order of –I effect of the given substituent is given by

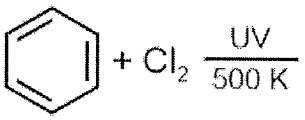
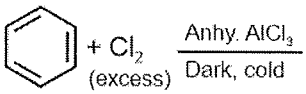
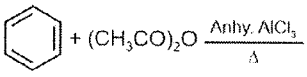
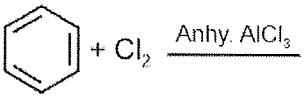
(1) – CN > –F > – OH

(2) –F > – OH > – CN

(3) –F > – CN > – OH

(4) –OH > – F > – CN

73. Match List-I with List-II

List-I		List-II
a.		(i) Acetophenone
b.		(ii) Chlorobenzene
c.		(iii) Hexachlorobenzene
d.		(iv) Gammaxane

Choose the correct answer from the options given below.

(1) a(iii), b-(ii), c-(iv), d-(i)

(2) a(i), b-(ii), c-(iii), d-(iv)

(3) a(ii), b-(iii), c-(iv), d-(i)

(4) a(iv), b-(iii), c-(i), d-(ii)

74. $CH_4 + O_2 \xrightarrow[\Delta]{Mo_2O_3} A + H_2O$

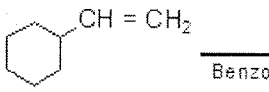
Product A is

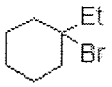
(1) CH_3OH

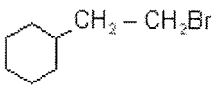
(2) HCHO

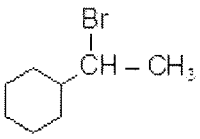
(3) HCOOH

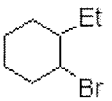
(4) CO_2

75.  (A) is (Major)

(1) 

(2) 

(3) 

(4) 

76. In the following reaction,

Acetic acid $\xrightarrow{NaOH(aq)} A \xrightarrow[\Delta]{Soda\ lime} B$

B is

(1) Ethene

(2) Methanol

(3) Ethanol

(4) Methane

77. 1.5 mole of n-hexane is mixed with 0.5 mole of n-pentane to form an ideal solution at 25°C. If the vapour pressure of n-hexane and n-pentane at 25°C are 150 mm and 200 mm of Hg respectively then the vapour pressure of the resulting solution in mm of Hg will be

(1) 215.5

(2) 250

(3) 162.5

(4) 175

78. Ethylene glycol is an antifreezing agent. The amount of ethylene glycol to be added to 2 kg of water to prevent it freezing at –5.58°C is (K_f of water = 1.86 K kg mol^{-1} and molar mass of ethylene glycol = 62 $gmol^{-1}$)

(1) 150 g

(2) 75 g

(3) 252 g

(4) 372 g

79. Which of the following gases is least soluble in water at 298 K?

Gases	K_H/kbar
Argon	40.3
Carbon dioxide	1.67
Formaldehyde	1.83×10^{-5}
Methane	0.413

The **correct** answer is

(1) Argon

(2) Carbon dioxide

(3) Formaldehyde

(4) Methane

80. Given below are two statements one is assertion (A) other is reason (R)

Assertion (A): Aquatic species are more comfortable in cold water than in warm water.

Reason (R): Solubility of oxygen in water decreases with increase in temperature.

In the light of above statements choose the correct answer.

(1) Both assertion and reason are correct statements, and reason is the correct explanation of the assertion.

(2) Both assertion and reason are correct statements, but reason is not the correct explanation of the assertion.

(3) Assertion is correct, but reason is wrong statement.

(4) Assertion is wrong, but reason is correct statement.

81. Azeotropic composition of nitric acid and water mixture is (by mass)

(1) 50% nitric acid and 50% water

(2) 1% nitric acid and 99% water

(3) 99% nitric acid and 1% water

(4) 68% nitric acid and 32% water

82. Products of electrolysis of aqueous solution of Na_2SO_4 using Pt electrodes are

(1) SO_2 and O_2

(2) H_2 and O_2

(3) H_2O_2 and SO_2

(4) H_2O_2 and H_2

83. If equilibrium constant of the given galvanic cell is 1×10^{100} at 25°C then standard electrode potential of the cell will be

$M(s) + N^{2+}(aq) \rightarrow M^{2+}(aq) + N(s)$

(1) 1.5 V

(2) 2.96 V

(3) 1.85 V

(4) 2.50 V

84. Match List-I with List-II

	List-I		List-II
a.	$Fe_4[Fe(CN)_6]_3.xH_2O$	(i)	Yellow
b.	$(NH_4)_3PO_4.12MoO_3$	(ii)	Prussian Blue
c.	$[Fe(CN)_5NOS]^{4-}$	(iii)	Violet
d.	$[Fe(SCN)]^{2+}$	(iv)	Blood red

Choose the correct answer from the options given below.

(1) a(ii), b(iii), c(iv), d(i)

(2) a(i), b(iii), c(ii), d(iv)

(3) a(ii), b(i), c(iii), d(iv)

(4) a(iii), b(i), c(iv), d(ii)

85. IUPAC name of the given compound is



(1) 3-Methyl-1,1-diethylcyclopentane

(2) 1,1-Diethyl-3-methylcyclopentane

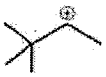
(3) 1-Methyl-3,3-diethylcyclopentane

(4) 3,3-Diethyl-1-methylcyclopentane

SECTION-B

86. Which of the following is the most stable carbocation?

(1) 

(2) 

(3) 

(4) 

87. In the Kjeldahl's method for the estimation of nitrogen present in a soil sample, ammonia evolved from 0.8 g of sample was neutralized by 20 ml of 1 M H_2SO_4 solution. The percentage of nitrogen in the soil is

(1) 35%

(2) 40%

(3) 50%

(4) 70%

88. Which among the following is not planar in structure?

(1) $CH_2 = CH - CN$

(2) $CH_2 = C = CH_2$

(3) $CH_2 = CH - CHO$

(4) $CH_2 = CH - CH = CH_2$

89. $NaNH_2$ is used to distinguish

(1) 2-butene and 1-butene

(2) 2-butene and butane

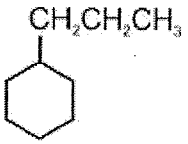
(3) 2-butyne and butane

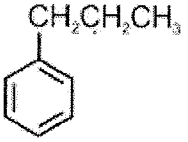
(4) 2-butene and 1-butyne

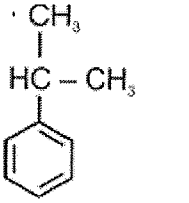
90. In the reaction sequence

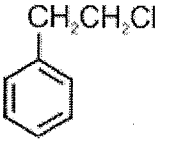
$$CaC_2 \xrightarrow{H_2O} A \xrightarrow[873\ K]{\text{Red hot Fe tube}} B \xrightarrow[\text{Anhy. } AlCl_3]{CH_3CH_2CH_2Cl} C \text{ (Major)}$$

C is

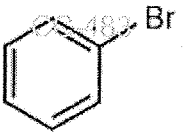
(1) 

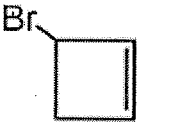
(2) 

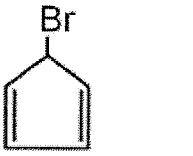
(3) 

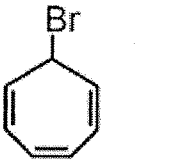
(4) 

91. Select the molecule which will give fastest precipitation reaction with $AgNO_3$ solution?

(1) 

(2) 

(3) 

(4) 

92. At 2 atm pressure, mole of $O_2(g)$ dissolved in 10 moles water at 293 K is (Henry's Law constant for $O_2(g)$ in water is 0.3×10^5 atm)

(1) 6.67×10^{-3}

(2) 6.67×10^{-5}

(3) 6.67×10^{-4}

(4) 6.67×10^{-6}

93. If molality of NaOH in an aqueous solution is 8m and the density of the solution is 1.5 g mL^{-1} then the molarity of NaOH solution will be

(1) 7.5M

(2) 7.0M

(3) 9.1M

(4) 8M

94. Consider the following statements.

(i) The solutions which show a large positive deviation from Raoult's law form minimum boiling azeotrope.

(ii) Two solutions having same osmotic pressure at a given temperature are called isotonic solutions.

(iii) K_f is independent of solute concentration.

The correct statement(s) is/are

(1) (i) and (ii) only

(2) (ii) and (iii) only

(3) (i), (ii) and (iii)

(4) (ii) only

95. An ideal solution is prepared by mixing of two volatile liquids A and B. If the vapour pressure of the solution is 180 mm of Hg, then the mole fraction of B in vapour phase is

$\left(P_A^\circ = 150 \text{ mm of Hg and } P_B^\circ = 200 \text{ mm of Hg}\right)$

(1) 0.67

(2) 0.34

(3) 0.75

(4) 0.50

96. Equilibrium constant for the reaction

$Ni(s) + 2\,Ag^{\oplus}(aq) \rightleftharpoons Ni^{2+}(aq) + 2\,Ag(s)$

at 298 K approximately is

[Given $E^\circ_{Ni^{2+}/Ni} = -0.25$ V, $E^\circ_{Ag^{\oplus}/Ag} = +0.80$ V]

(1) $10^{35.5}$

(2) $10^{18.5}$

(3) $10^{30.5}$

(4) $10^{14.5}$

97. Standard oxidation potentials of the half reaction are given below

$Fe^{2+} \rightarrow Fe^{3+} + e^-$, $E^o = -0.77$ V

$2I^- \rightarrow I_2 + 2e^-$, $E^o = -0.536$ V

$Mn \rightarrow Mn^{2+} + 2e^-$, $E^o = +1.18$ V

$Zn \rightarrow Zn^{2+} + 2e^-$, $E^o = +0.76$ V

. The strongest reducing agent among the following is

(1) Fe^{2+}

(2) Mn

(3) I^-

(4) Zn

98. Consider the following galvanic cell.

$Mg(s) \mid Mg^{2+}(2 \times 10^{-4}\ M) \parallel Ag^+(2 \times 10^{-2}\ M) \mid Ag(s)$

If $E^{\circ}_{Mg^{2+}/Mg} = -2.36$ V and $E^{\circ}_{Ag^+/Ag} = 0.80$ V then emf of the cell will be

(1) 4.32 V

(2) 2.44 V

(3) 1.52 V

(4) 3.17 V

99. Given

(i) $A^{2+} + 2e^- \rightarrow A$, $E^{\circ} = 0.2$ V

(ii) $A^{2+} + e^- \rightarrow A^+$, $E^{\circ} = 0.1$ V

Electrode potential, E° for the reaction

$A^+ + e^- \rightarrow A$, will be

(1) 0.1 V

(2) 0.2 V

(3) 0.3 V

(4) 0.4 V

100. An aqueous solution of CH_3COONa on electrolysis gives which hydrocarbon?

(1) CH_4

(2) C_2H_6

(3) C_2H_4

(4) C_2H_2

BOTANY

SECTION-A

101. Select the **incorrect** statement among the following.

(1) Pollen germination and pollen tube growth are regulated by chemical components of the pollen interacting with those of pistil

(2) Some reptiles are reported as potential pollinators in some plant species.

(3) There are certain insects that consume nectar or pollens without bringing about pollination are called pollen robbers

(4) In castor and maize, transfer of pollen grains from anther to stigma of another flower of the same plant is prevented.

102. In how many among the following plants mature seeds retain a part of endosperm as it is **not** completely used up during embryo development?

Barley, Castor, Pea, Wheat, Maize, Groundnut

(1) Two

(2) Four

(3) Three

(4) Five

103. Select the **correctly** matched one.

(1) Hypocotyl – It is the portion of embryonal axis above the level of cotyledons which terminates with root tip

(2) Coleoptile – It is an undifferentiated sheath which encloses root cap and radical

(3) Coleorhiza – A hollow foliar structures which encloses few leaf primordia

(4) Epicotyl – It is the portion of the embryonal axis above the level of attachment of the cotyledons

104. Which among the following options is **incorrect** regarding seeds?

(1) It is the basis of agriculture

(2) It is the final product of sexual reproduction

(3) It cannot be formed without fertilisation

(4) Some of them can remain alive for thousands of years

105. In angiosperms, triple fusion differs from syngamy as the former

(1) Results in the formation of diploid cell, the zygote.

(2) Involves fusion of a male gamete with two polar nuclei.

(3) Involves formation of primary endosperm cell which gradually develops into embryo.

(4) Involves fusion of three diploid nuclei.

106. Read the following statements of **Assertion (A)** and **Reason (R)** and choose the **correct** option.
Assertion (A): In some plants, such as *Viola*, *Oxalis* and *Commelina*, all flowers are invariably autogamous as there is no chance of cross pollen landing on the stigma.
Reason (R): *Viola*, *Oxalis* and *Commelina* produces flowers in which anther and stigma lie close to each other and flowers do not open at all.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

107. The egg apparatus of embryo sac in angiosperm consists of

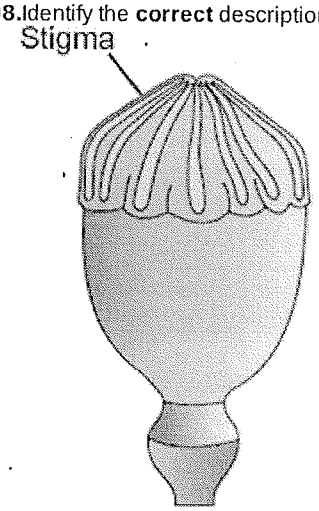
(1) One egg cell, two synergids, three antipodals and two polar nuclei

(2) One egg cell and two synergids

(3) Three antipodal cells and two polar nuclei

(4) Filiform apparatus, one egg cell and a central cell

108. Identify the **correct** description about the given figure



(1) A multicapellary apocarpous gynoecium of *Michelia*

(2) A diagrammatic view of a ovary having distinct carpels

(3) A multicarpellary, syncarpous pistil of *Papaver*

(4) A dissected flower of *Hibiscus* showing pistil

109. Male gametophytes of flowering plants remain preserved as fossil due to the presence of

(1) Tapetum

(2) An organic material in exine

(3) Epidermal layer

(4) Pectin in intine

110. Choose the **correct** sequence of events involved in the formation of embryo sac in flowering plants.
A. Migration of three nuclei towards each pole
B. Only one megaspore is functional while other degenerate.
C. Formation of cell wall around nuclei results in seven celled structure.
D. Reductional division in the megaspore mother cell.
E. Three sequential mitotic free nuclear division in functional megaspore.
Choose the **correct** option

(1) B → E → D → C → A

(2) D → E → B → A → C

(3) D → B → E → A → C

(4) B → D → E → C → A

111. Regarding pollination in *Vallisneria*, choose the **correct** statement?

(1) The flower emerges above the level of water and get pollinated by insects.

(2) Male flowers or pollen grains are carried by water currents to female flower at surface of water.

(3) Female flowers remain submerged in water and pollen grains are released inside the water.

(4) Pollens are long ribbon like and they float actively inside the water

112. Given below are two statements.
Statement I: In pollen grain, the cell with abundant food reserve and large irregularly shaped nucleus divides mitotically to give rise to two male gametes.
Statements II: In some members of Rosaceae, Leguminoseae and Solanaceae pollen grains remain viable for months.
In the light of above statements choose the **correct** answer from the options given below.

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

113. Before entering the respiratory pathway the fatty acids would be degraded to

(1) Pyruvic acid

(2) Dihydroxy acetone phosphate

(3) Acetyl CoA

(4) α-ketoglutaric acid

114. Which among the following processes occur in aerobic conditions.
i. Glycolysis
ii. Krebs' cycle
iii. Oxidative decarboxilation
iv. Alcoholic fermentation

(1) i, ii and iii only

(2) ii and iii only

(3) i and ii only

(4) iii and iv only

115. Match the list I with list II and choose the **correct** option.

	List I		List II
a.	Complex III	(i)	Involved in two processes associated with aerobic respiration
b.	Complex I	(ii)	Produces ATP from ADP
c.	Complex V	(iii)	Cytochrome bc_1
d.	Complex II	(iv)	NADH dehydrogenase

(1) a(ii), b(iv), c(i), d(iii)

(2) a(iii), b(iv), c(i), d(ii)

(3) a(iii), b(i), c(iv), d(ii)

(4) a(iii), b(iv), c(ii), d(i)

116. Read the following w.r.t. auxin.
i. It has been extensively used in agricultural practices.
ii. It promotes fruit and leaf drop at early stages of plant.
iii. It was first isolated from tip of coleoptile of oat.
iv. It is composed of indole compound.
The **correct** ones are

(1) i, iii and iv only

(2) i and iv only

(3) ii and iii only

(4) All i, ii, iii and iv

117. By the application of which PGRs, farmers can left fruits on tree longer so as to extend its market period?

(1) Gibberellin

(2) Auxin

(3) Cytokinin

(4) Abscisic acid

118. Read the following statements and choose the **correct** option.
Statement A: In C_4 plants, the primary CO_2 acceptor is a three carbon molecule and is present in bundle sheath cell.
Statement B: During C_4 cycle of photosynthesis, the acid containing four carbon is formed in mesophyll cells.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statements A and B are correct

(4) Both statements A and B are incorrect

119. The phases of growth during which growth progresses rapidly, *i.e.*, riches to maximum is

(1) Stationary phase

(2) Log phase

(3) Lag phase

(4) Senescence

120. State the following statements as True (T) or false (F) and choose the **correct** option.
A: E. Kurosawa found that the foolish seedling disease was caused by a virus *Gibberella fujikuroi*.
B: Kinetin was the first natural cytokinin which was isolated from unripe maize grains or kernel.

	A	B
(1)	T	F
(2)	F	F
(3)	T	T
(4)	F	T

(1) (1)

(2) (2)

(3) (3)

(4) (4)

121. Leaves of rice seedling remain out of water due to

(1) Increase in indole compound that leads to rapid division of the cells

(2) Increased ABA concentration that increases the length of the stem.

(3) Gaseous phytohormone that induces rapid elongation of internode and petiole.

(4) Increased concentration of cytokinin that inhibits the growth of lateral bud

122. Oxidative phosphorylation is

(1) Removal of phosphate group from ATP

(2) Oxidation of phosphate group in ATP

(3) Direct formation of ATP by transfer of phosphate group from a substrate to ADP

(4) Formation of ATP by energy released from electrons removed during substrate oxidation

123. In both glycolysis and tricarboxylic acid cycle the

(1) Decarboxylation reaction takes place

(2) Oxygen requirement is obligatory

(3) First intermediate of the processes is a 6C molecule

(4) Only one substrate level phosphorylation reaction occurs

124. In C_4 plants, the cells which form several layers around the vascular bundle are characterised by having
i. Thin walls which are permeable to gaseous exchange.
ii. Large numbers of chloroplasts.
iii. Large intercellular spaces.
The **correct** one(s) is/are

(1) ii and iii only

(2) i only

(3) All i, ii and iii

(4) ii only

125. Who confirmed the release of a volatile substance from ripened oranges that hastened the ripening of stored unripened bananas?

(1) H.H. Cousins

(2) E. Kurosawa

(3) F. Skoog

(4) Miller

126. Consider the following statements and choose the **correct** option.
A. Action spectrum of photosynthesis corresponds closely to absorption spectra of chlorophyll *b* which shows that chlorophyll *a* is chief pigment associated with photosynthesis.
B. Most of the photosynthesis takes place in the blue and red region of the visible spectrum.
C. All antennae pigment absorb light of same wavelength and make the photosynthesis efficient.
D. Carotenoids is an accessory pigment and protect chlorophyll *a* from photo oxidation.

(1) Statements B and C are incorrect

(2) Statements A and C are incorrect

(3) Statements B, C and D are correct

(4) Statements A and D are incorrect

127. Kranz anatomy of leaves are observed in

(1) Bell pepper

(2) Tomato

(3) Sorghum

(4) Wheat

128. All of the following are internal plant factor affecting photosynthesis, **except**

(1) Amount of chlorophyll

(2) Temperature

(3) Orientation of leaves

(4) Mesophyll cells of leaves

129. A plant growth regulator which is also known as stress hormone

(1) Breaks dormancy of buds

(2) Promotes stem elongation in submerged plant

(3) Is a derivative of carotenoids

(4) Causes quicker maturity in the juvenile conifers

130. Heterophyllous development due to environment is observed in

(1) Coriander

(2) Buttercup

(3) Larkspur

(4) Cotton

131. When tripalmitin is used as respiratory substrate, then expected value of respiratory quotient would be

(1) 1.4

(2) 0.9

(3) 1

(4) 0.7

132. Citric acid cycle

(1) Involves enzymes that are present only in soluble state in mitochondrial matrix

(2) Serves as a common oxidative pathway for fats, carbohydrates and proteins.

(3) Is confined to the cytosol of the eukaryotic cell.

(4) Is involved in the formation of two molecules of acetyl CoA from one pyruvic acid.

133. For trapping light in leaves, light harvesting complex in plants contain

(1) All the pigments except chlorophyll *a*

(2) Chlorophyll *a* only

(3) Many accessory pigments along with chlorophyll *a*

(4) Only chlorophyll *a* and chlorophyll *b*

134. Which among the following processes is **not** common between C_3 and C_4 plants ?

(1) Glycolysis

(2) Respiration

(3) Calvin cycle

(4) Photorespiration

135. Choose the **incorrect** statement w.r.t. early historical account of photosynthesis.

(1) Jan Ingenhousz showed that in the presence of sunlight, only the green parts of the plants could release oxygen.

(2) Cornelius van Niel concluded that the plant restore air whatever the breathing mouse and burning candle remove

(3) Julius von Sachs found that green parts of the plant form glucose

(4) T.W. Engelmann performed experiment using *Cladophora*

SECTION-B

136. Select the common feature between wind and water pollinated flowers.

(1) Pollen grains are protected from wetting by a mucilaginous covering

(2) Flowers secrete foul odour to attract pollinators

(3) The flowers are large and fragrant

(4) Flowers are not very colourful and do not produce nectar

137. Which among the following is **incorrect** w.r.t. conduction of artificial hybridisation for plants?

a. There is no need of emasculation in the female parent having unisexual flower.

b. The plant producing only female flowers need not to be bagged in their bud stage.

c. Emasculated flowers are needed to be bagged immediately after cross pollination.

d. Bisexual flowers, showing maturation of gynoecium before the androecium, are never selected for cross pollination.

e. Bagging of emasculated flowers after removal of anthers.

(1) b and d only

(2) a, c and d only

(3) a, b and d only

(4) b and e only

138. Both autogamy and geitonogamy are prevented in

(1) Wheat

(2) Papaya

(3) Castor

(4) Maize

139. The megasporangium is a small structure attached to a placenta by means of a stalk called

(1) Hilum

(2) Integument

(3) Funicle

(4) Embryo sac

140. Match the column I with column II and choose the **correct** option.

	Column I		Column II
a.	Derivatives of carotenoids	i.	Inhibit the growth of lateral buds
b.	Indole compounds	ii.	Are used to speed up the malting process in brewing industry
c.	Adenine derivatives	iii.	Play an important role in seed development and dormancy
d.	Derivatives of terpenes	iv.	Are synthesized in regions where rapid cell division occur

(1) a(iii), b(iv), c(i), d(ii)

(2) a(iii), b(i), c(iv), d(ii)

(3) a(ii), b(iv), c(i), d(iii)

(4) a(iii), b(i), c(ii), d(iv)

141. To boost the production of tea leaves, shoot tip leaves are removed. This is because

(1) Ethylene induces swelling of the axis and thus apical hook formation takes place

(2) Gibberellins delay senescence of leaves

(3) Effect of auxin is removed and growth of lateral bud is enhanced

(4) Auxin prevents abscission of mature leaves

142. The phytohormone which causes an increase in length of grapes stalk is

(1) Cytokinin

(2) ABA

(3) Auxin

(4) Gibberellin

143. At high light intensities C_3 plants show photosynthetic saturation

(1) At 360 µl L^{-1} of CO_2 concentration

(2) Only beyond 450 µl L^{-1} of CO_2 concentration

(3) Below 300 µl L^{-1} of CO_2 concentration

(4) Only at 400 µl L^{-1} of CO_2 concentration

144. In cyclic photophosphorylation

(1) Both photosystem I and photosystem II are involved

(2) Oxygen is not released

(3) External electron donor is required

(4) ATP along with NADPH is synthesised

145. In which of the following reactions NAD^+ is not reduced ?
i. Succinic acid → Malic acid.
ii. Pyruvic acid → Acetyl CoA.
iii. α-ketoglutaric acid → Succinic acid.
iv. Pyruvic acid → Lactic acid.
v. Glyceraldehyde-3-phosphate → 1,3-bisphosproglyceric acid.
vi. Phosphoenol pyruvate → Pyruvic acid
The **correct** ones are

(1) i, ii and iii

(2) iii, iv and v

(3) ii, iii and v

(4) i, iv and vi

146. The incomplete oxidation of glucose is achieved under anaerobic conditions by sets of reactions where pyruvic acid is converted to CO_2 and ethanol. This is achieved by the set of enzymes

(1) Pyruvic acid dehydrogenase and alcohol dehydrogenase

(2) Lactate dehydrogenase and pyruvate dehydrogenase

(3) Pyruvic acid decarboxylase and alcohol dehydrogenase

(4) Pyruvate decarboxylase and lactate dehydrogenase

147. The reduced ubiquinol is oxidised with the transfer of electrons to

(1) Cytochrome *c* via cytochrome bc_1 complex

(2) Fe-S via FAD

(3) Complex III via complex I

(4) Cytochrome *a* via copper

148. F_0 particle of ATP synthase

(1) Is a peripheral membrane protein complex

(2) Contains the site of ATP synthesis

(3) Passes electron from intermembrane space to matrix

(4) Form channel through which proton crosses the membrane

149. Read the following statements.
Assertion (A): Carboxylation leads to fixation of carbon dioxide into the stable organic compound intermediate.
Reason (R): Carboxylation is the most crucial step of the Calvin cycle.
In the light of above statements, choose the **correct** option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

150. The oxygenase activity of RuBisCO enzyme in C_3 plant leads to the formation of

(1) 2 molecules of 3C compound

(2) 1 molecule of 4C compound and one molecule of 2C compound

(3) 1 molecule of 3C compound and 1 molecule of 2C compound

(4) 1 molecule of 6C compound and 1 molecule of 3C compound

ZOOLOGY

SECTION-A

151. An animal which does not possess a pseudocoelom is

(1) Roundworm

(2) Hookworm

(3) Tapeworm

(4) Filarial worm

152. Complete the analogy w.r.t. phylum and its members.
Largest phylum : *Culex* : : Second largest phylum : _____
Select the **correct** option.

(1) *Dentalium*

(2) *Ophiura*

(3) *Nereis*

(4) *Echinus*

153. Select the correct match w.r.t. taxonomic position of Lamprey.

(1) Phylum : Vertebrata

(2) Sub phylum : Gnathostomata

(3) Division : Agnatha

(4) Super class : Cyclostomata

154. 'X' is a bird that cannot fly while 'Y' is a mammal that can fly. Identify 'X' and 'Y' respectively, from the options given below.

(1) *Struthio* ; *Macropus*

(2) *Neophron* ; *Macaca*

(3) *Aptenodytes* ; *Pteropus*

(4) *Corvus* ; *Equus*

155. Different groups of viruses can cause all of the following diseases, **except**

(1) Cancer

(2) AIDS

(3) Common cold

(4) Plague

156. Select the incorrect statement w.r.t. dependence.

(1) It is the tendency to manifest unpleasant withdrawal syndrome on discontinuing regular dose of drugs/alcohol.

(2) It is characterised by nausea, anxiety and sweating.

(3) It is physiological attachment to certain effects like euphoria and temporary feeling of well being.

(4) Medication, guidance, counselling and avoiding undue peer pressure can be helpful for treatment.

157. A major chemical substance present in tobacco stimulates adrenal medulla to release catecholamines into blood circulation. This chemical substance belongs to which of the following categories of secondary metabolites?

(1) Lectin

(2) Alkaloid

(3) Pigment

(4) Polymeric substance

158. The body of *Ctenoplana* bears _____ external rows of _____ comb plates, which help in locomotion.
Fill in the respective blanks by selecting the **correct** option.

(1) 4; Flagellated

(2) 4; Ciliated

(3) 8; Flagellated

(4) 8; Ciliated

159. **Assertion (A):** Hemichordata was earlier considered as a sub-phylum under the phylum Chordata.
Reason (R): A rudimentary structure in the collar region called stomochord was earlier considered to be nerve cord.
In the light of above statements, select the **correct** option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

160. All of the following are the properties of non-cancerous normal cells, **except**

(1) Contact inhibition

(2) Controlled cell growth

(3) Regulated cell differentiation

(4) Metastasis

161. A person was bitten by a snake. For quick immune response, the injection given to him against snake venom would contain

(1) Weakened pathogen

(2) Antigenic protein of pathogen

(3) Pre-formed antibodies

(4) Dead microbes

162. The disease 'X' is caused due to infection by *Trichophyton*. Choose the incorrect option w.r.t. 'X'.

(1) Characterised by dry, scaly lesions on skin, nails and scalp

(2) Pathogen causing the disease 'X' thrives in the skin folds of groin region and in between toes

(3) This disease spreads by using clothes, towels or combs of an infected person

(4) Characterised by stools with excess mucus and blood clots

163. Consider the following features

(a) Diploblastic
(b) Absence of statocyst
(c) Presence of chitinous exoskeleton
(d) Exhibit bioluminescence
(e) Presence of three pairs of appendages

Which of the following options is true w.r.t. scorpion from the features given above?

(1) a, b, c, d only

(2) b, c, e only

(3) a, c, d, e only

(4) b and c only

164. Which of the following characteristics is shared by both mammals and birds, differentiating them completely from reptiles and amphibians ?

(1) Presence of a dorsal hollow nerve cord

(2) Homeothermy

(3) Direct development

(4) 4-chambered heart

165. Comprehend the given statements.

(a) Antibodies against cancer specific drugs can be used for treatment of cancer.
(b) Macrophages act as HIV factory by constantly destroying and killing HIV.
(c) MALT constitutes about 50% of the lymphoid tissue in human body.

Select the option representing correct statement(s) only.

(1) (a) and (b)

(2) (b) only

(3) (c) and (a)

(4) (c) only

166. Choose the correct match w.r.t. class and the unique feature of its members.

(1) Reptilia - Tympanum represents ear

(2) Aves - Presence of feathers on body

(3) Amphibia - Scales on body

(4) Mammalia - Functional mammary glands in all the members

167. Select the odd one w.r.t. poisonous snakes.

(1) *Naja*

(2) *Bangarus*

(3) *Chelone*

(4) *Vipera*

168. Which of the following characteristics is true for most of the members of both the respective phyla to which *Culex* and *Dentalium* belong to?

(1) They possess segmented body

(2) Their body is covered by calcareous shell

(3) They possess muscular foot

(4) They exhibit indirect development

169. The enzyme that has ability to convert ssRNA of retrovirus into ssDNA is

(1) Reverse transcriptase

(2) Restriction endonuclease

(3) RNA polymerase

(4) Ligase

170. Interferon, an antiviral protein, is included under which of the following types of barriers of innate immunity?

(1) Cellular

(2) Physical

(3) Physiological

(4) Cytokine

171. **Assertion (A):** Colostrum secreted by mother during initial days of lactation provides passive immunity to baby.
Reason (R): Colostrum abundantly contains antibodies IgE, which are transferred from mother to the foetus through placenta.
In the light of above statements, choose the **correct** option.

(1) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(2) Both (A) and (R) are true and (R) is the correct explanation of (A).

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

172. A similarity between *Sycon*, *Taenia* and *Hirudinaria* is that all of them

(1) Are monoecious

(2) Show indirect development

(3) Show external fertilisation

(4) Exhibit extracellular digestion of food

173. _____ exhibits metagenesis while _____ exhibits true metamerism.
Fill in the respective blanks by selecting the **correct** option.

(1) *Obelia* ; *Nereis*

(2) *Adamsia* ; *Sycon*

(3) *Pleurobrachia* ; *Chaetopleura*

(4) *Ctenoplana* ; *Echinus*

174. Total number of peptide chain(s) in a monomeric antibody molecule is

(1) One

(2) Two

(3) Three

(4) Four

175. Select the correct set of diseases caused by bacterial pathogens.

(1) Diphtheria, Common cold, Amoebiasis

(2) Tetanus, Pneumonia, Diphtheria

(3) Elephantiasis, Typhoid, AIDS

(4) Tetanus, Malaria, Ringworms

176. Consider the given statements:
Statement A: All mammals have four chambered heart but all animals with 4 chambered heart are not mammals.
Statement B: All mammals are viviparous but all animals exhibiting viviparity are not mammals.
Select the correct option.

(1) Both statements A and B are incorrect

(2) Both statements A and B are correct

(3) Only statement A is correct

(4) Only statement B is correct

177. Select the correct match w.r.t. disease and its prominent symptom.

(1)	Ascariasis	–	Chronic inflammation of lower limbs or gonads
(2)	Filariasis	–	Intestinal perforations
(3)	Ringworm	–	Chills and recurring high fever
(4)	Pneumonia	–	Finger nails turn grey to bluish in severe case

(1) (1)

(2) (2)

(3) (3)

(4) (4)

178. Match column I and column II w.r.t. components belonging to the same category of barriers of innate immunity.

	Column-I		**Column-II**
a.	Skin	(i)	Acid in stomach
b.	Tears	(ii)	Monocytes
c.	PMNL	(iii)	Mucus coating of GIT

Select the correct option.

(1) a(iii), b(i), c(ii)

(2) a(ii), b(iii), c(i)

(3) a(ii), b(i), c(iii)

(4) a(iii), b(ii), c(i)

179. Which of the following is not a characteristic feature of the class to which *Pavo* belongs to?

(1) 4-chambered heart in the members

(2) Presence of sweat glands and oil glands throughout the body

(3) Hindwings modified for walking, swimming or clasping the tree branches

(4) Pneumatic long bones

180. The water vascular system in *Antedon* does not help in/to

(1) Locomote from one place to another

(2) Capture of food

(3) Gaseous exchange

(4) Fertilisation

181. Which of the following is **incorrect** for the spiny bodied animals with endoskeleton of calcareous ossicles?

(1) All are exclusively marine

(2) Have mouth on ventral side and anus on dorsal side

(3) The anterior head region has sensory tentacles

(4) Segmentation is absent in their body

182. A person shows symptoms of a severe allergic reaction after eating peanuts. What should be given to the patient for immediate treatment in such case?

(1) Anti-retroviral drugs

(2) Steroids and histamine

(3) Adrenaline and steroids

(4) Histamine and serotonin

183. Which of the following statements is correct about T-lymphocytes?

(1) They are directly responsible for humoral immune response

(2) They directly produce antibodies in response to antigens

(3) They are a part of cytokine barrier of innate immunity

(4) They originate from bone marrow but mature in thymus

184. The class to which *Exocoetus* belongs to differs from the class to which *Pristis* belongs, in all of the following aspects, **except**

(1) Number of heart chambers

(2) Structure that regulates buoyancy

(3) Location of mouth

(4) Structure that covers gill slits

185. How many drugs given in the box below have hallucinogenic properties?

Heroin, Marijuana, Morphine, Ganja, Charas

Select the correct option.

(1) Two

(2) Three

(3) Four

(4) Five

SECTION-B

186. In an animal belonging to the 'X', notochord is present only in larval tail while in the animal belonging to the 'Y', it is persistent throughout their life extending from head to tail region.

Select the correct option w.r.t. 'X' and 'Y'.

	'X'	'Y'
(1)	Subphylum–Urochordata	Division–Cephalochordata
(2)	Division–Protochordata	Division–Urochordata
(3)	Class–Cyclostomata	Sub-Class–Protochordata
(4)	Subphylum–Urochordata	Subphylum–Cephalochordata

(1) (1)

(2) (2)

(3) (3)

(4) (4)

187. The malarial parasite reproduces asexually in

(1) Both hepatocytes and RBCs of human host

(2) Salivary glands and gut of female *Anopheles* mosquito

(3) Human RBCs and gut of female *Aedes* mosquito

(4) Human hepatocytes and female *Anopheles* mosquito's salivary glands

188. Complete the analogy w.r.t. location of receptors of given drugs in human body by selecting the correct option.

Marijuana : Brain :: Heroin: _____

(1) Brain only

(2) CNS and GIT

(3) Heart only

(4) GIT only

189. All of the following are freshwater bony fishes, **except**

(1) *Labeo*

(2) *Catla*

(3) *Clarias*

(4) *Hippocampus*

190. The type of immune response which is majorly responsible for graft rejection is mediated by

(1) Antibodies

(2) B-Lymphocytes

(3) T-Lymphocytes

(4) PMNLs

191. **Assertion (A):** A person suffering from AIDS become more susceptible to bacterial, fungal and viral infections.
Reason (R): HIV enters helper T lymphocytes, replicates and produces progeny virus.
In the light of the above statements, select the **correct** option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

192. Anamnestic immune response differs from primary immune response as

(1) Former is produced during the first encounter with pathogen

(2) Latter is of high intensity than former

(3) Former elicits highly intensified response

(4) Latter is more rapid and vigorous than former

193. Column I represents a particular feature and column II represents members of the phyla in which that feature appeared for the first time. Match column I with column II and select the correct option.

	Column I		Column II
a.	Complete digestive system	(i)	*Planaria*
b.	Bilateral symmetry	(ii)	*Nereis*
c.	Closed circulatory system	(iii)	*Ascaris*
d.	Respiratory system	(iv)	*Anopheles*

(1) a(i), b(iii), c(iv), d(ii)

(2) a(iv), b(i), c(ii), d(iii)

(3) a(iii), b(ii), c(i), d(iv)

(4) a(iii), b(i), c(ii), d(iv)

194. How many animals given in the box below are oviparous, poikilotherms and exhibit external fertilisation?

Pterophyllum, *Bangarus*, *Columba*, *Bufo*, *Pristis*

Select the correct option.

(1) Three

(2) Two

(3) Five

(4) One

195. Select the odd one w.r.t. animals exhibiting indirect development.

(1) *Bombyx*

(2) *Fasciola*

(3) *Balanoglossus*

(4) *Pheretima*

196. AIDS was first reported in the year

(1) 1951

(2) 1961

(3) 1971

(4) 1981

197. Select the incorrect match.

	Animal	Structure	Function
(1)	*Pheretima*	Longitudinal and circular muscles	Locomotion
(2)	*Loligo*	Radula	Feeding
(3)	*Laccifer*	Statocyst	Sensory organ
(4)	*Pennatula*	Cnidoblast	Defense

(1) (1)

(2) (2)

(3) (3)

(4) (4)

198. Consider the given statements:
Statement A: Amphetamines are normally used as medicines to help patients cope with depression and insomnia.
Statement B: Smoking increases the concentration of haembound oxygen in the body of humans.
Select the correct option.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Only statement A is incorrect

(4) Only statement B is incorrect

199. Select the option representing correct point of difference between chordates and non-chordates.

	Chordates	Non-chordates
(1)	Dorsal heart	Ventral heart
(2)	CNS is dorsal, solid and single	CNS is ventral, hollow and double
(3)	Post anal tail present	Post anal tail absent
(4)	Larynx perforated by gill slits	Gill slits are absent

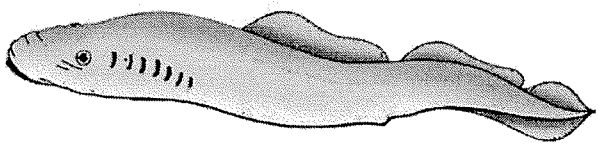
(1) (1)

(2) (2)

(3) (3)

(4) (4)

200. Identify the member of sub-phylum Vertebrata represented in the given figure.



Select the incorrect feature w.r.t. it

(1) Body is devoid of unpaired fins

(2) Cartilaginous cranium and vertebral column

(3) Closed circulation with ventral heart

(4) Exhibit anadromous migration

27-11-2024

Code-A

Phase-1_OYM

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NCERT BOOSTER TEST SERIES

(*for* NEET-2025)

Test - 1

Answers

1. (3)	41. (2)	81. (4)	121. (3)	161. (3)
2. (2)	42. (2)	82. (2)	122. (4)	162. (4)
3. (2)	43. (2)	83. (2)	123. (3)	163. (4)
4. (2)	44. (4)	84. (3)	124. (4)	164. (2)
5. (3)	45. (4)	85. (2)	125. (1)	165. (4)
6. (4)	46. (2)	86. (1)	126. (2)	166. (2)
7. (1)	47. (4)	87. (4)	127. (3)	167. (3)
8. (2)	48. (1)	88. (2)	128. (2)	168. (4)
9. (1)	49. (2)	89. (4)	129. (3)	169. (1)
10. (3)	50. (1)	90. (3)	130. (2)	170. (4)
11. (2)	51. (2)	91. (4)	131. (4)	171. (3)
12. (1)	52. (3)	92. (3)	132. (2)	172. (1)
13. (4)	53. (3)	93. (3)	133. (3)	173. (1)
14. (3)	54. (2)	94. (3)	134. (4)	174. (4)
15. (4)	55. (4)	95. (1)	135. (2)	175. (2)
16. (3)	56. (1)	96. (1)	136. (4)	176. (3)
17. (4)	57. (3)	97. (2)	137. (1)	177. (4)
18. (1)	58. (3)	98. (4)	138. (2)	178. (1)
19. (4)	59. (4)	99. (3)	139. (3)	179. (2)
20. (1)	60. (1)	100. (2)	140. (2)	180. (4)
21. (4)	61. (2)	101. (4)	141. (3)	181. (3)
22. (2)	62. (2)	102. (2)	142. (4)	182. (3)
23. (3)	63. (1)	103. (4)	143. (2)	183. (4)
24. (4)	64. (4)	104. (3)	144. (2)	184. (1)
25. (1)	65. (4)	105. (2)	145. (4)	185. (2)
26. (2)	66. (1)	106. (4)	146. (3)	186. (4)
27. (3)	67. (2)	107. (2)	147. (1)	187. (1)
28. (1)	68. (2)	108. (3)	148. (4)	188. (2)
29. (4)	69. (4)	109. (2)	149. (2)	189. (4)
30. (4)	70. (1)	110. (3)	150. (3)	190. (3)
31. (4)	71. (1)	111. (2)	151. (3)	191. (2)
32. (1)	72. (1)	112. (2)	152. (1)	192. (3)
33. (3)	73. (4)	113. (3)	153. (3)	193. (4)
34. (3)	74. (2)	114. (2)	154. (3)	194. (2)
35. (3)	75. (2)	115. (4)	155. (4)	195. (4)
36. (2)	76. (4)	116. (2)	156. (3)	196. (4)
37. (4)	77. (3)	117. (1)	157. (2)	197. (3)
38. (1)	78. (4)	118. (2)	158. (4)	198. (2)
39. (3)	79. (1)	119. (2)	159. (3)	199. (3)
40. (2)	80. (1)	120. (2)	160. (4)	200. (1)